CargoFlow

Working capital that releases only when the cargo's own evidence says it should.



Evidence-gated trade finance, settled in Paxos USDG on Robinhood Chain.

Production build live on Robinhood Chain Testnet. Mainnet release next.

Colosseum Crypto World's Fair · Robinhood Chain track · October 2026

`cargoflow.adoranto737.workers.dev` · `github.com/LSUDOKO/CargoFlow`

TRADE FINANCE TODAY

A lender sees paperwork, not the container.

So it lends blind, or does not lend at all. The exporter waits about two months for cash it has already earned, while the cargo is at risk in transit.



Daniel, the financier, holds the documents. The reefer stays out of sight.

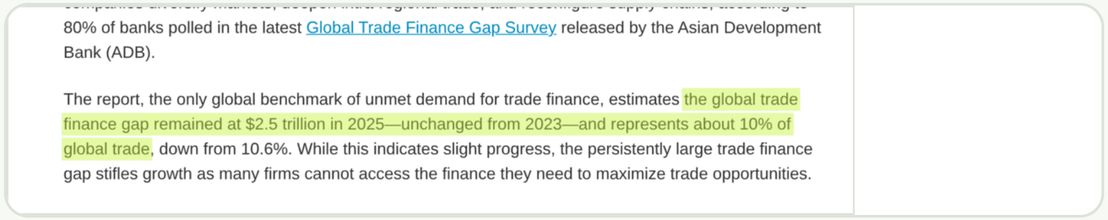
Atradius Payment Practices Barometer India 2025: average B2B payment terms of 52 days

SOURCED, HIGHLIGHTED ON THE ORIGINAL PAGES

The gap is large, and it is documented.

$2.5T

global trade finance gap in 2025 (ADB)

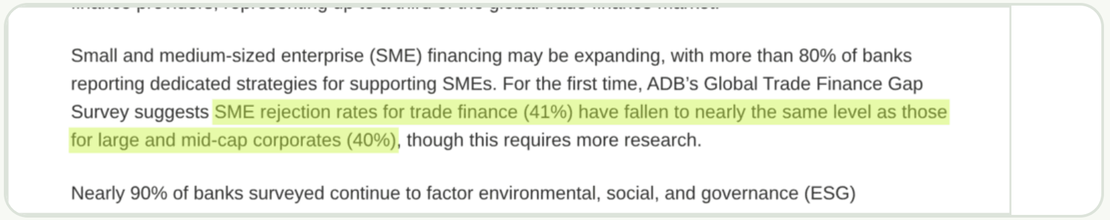


80% of banks polled in the latest Global Trade Finance Gap Survey released by the Asian Development Bank (ADB).

The report, the only global benchmark of unmet demand for trade finance, estimates the global trade finance gap remained at $2.5 trillion in 2025—unchanged from 2023—and represents about 10% of global trade, down from 10.6%. While this indicates slight progress, the persistently large trade finance gap stifles growth as many firms cannot access the finance they need to maximize trade opportunities.

41%

of SME trade finance applications rejected (ADB)

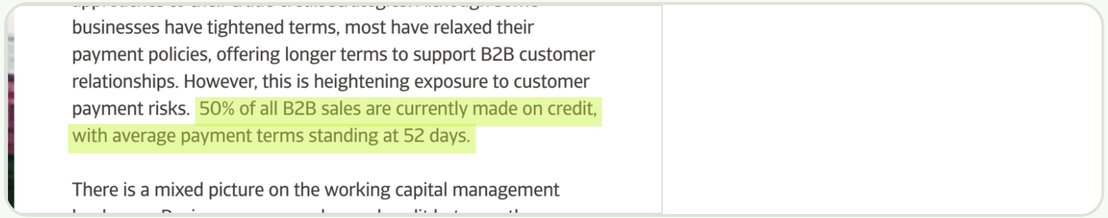


Small and medium-sized enterprise (SME) financing may be expanding, with more than 80% of banks reporting dedicated strategies for supporting SMEs. For the first time, ADB's Global Trade Finance Gap Survey suggests SME rejection rates for trade finance (41%) have fallen to nearly the same level as those for large and mid-cap corporates (40%), though this requires more research.

Nearly 90% of banks surveyed continue to factor environmental, social, and governance (ESG)

52 days

average B2B payment terms in India (Atradius)

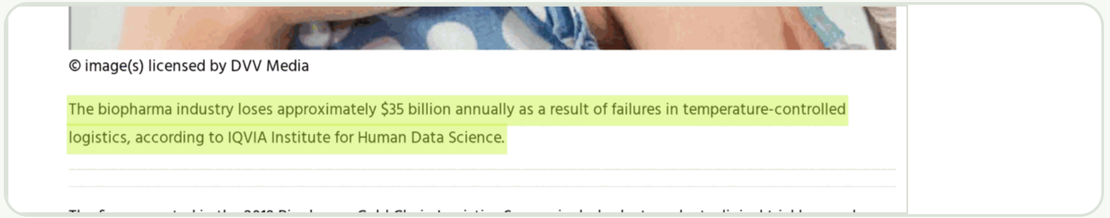


businesses have tightened terms, most have relaxed their payment policies, offering longer terms to support B2B customer relationships. However, this is heightening exposure to customer payment risks. 50% of all B2B sales are currently made on credit, with average payment terms standing at 52 days.

There is a mixed picture on the working capital management

~$35B

lost by biopharma each year to cold-chain failures (IQVIA)



© image(s) licensed by DVV Media

The biopharma industry loses approximately $35 billion annually as a result of failures in temperature-controlled logistics, according to IQVIA Institute for Human Data Science.

ADB, 15 Jan 2026 citing the ADB Global Trade Finance Gap Survey · Atradius India 2025, p. 3 · Air Cargo News, 26 Jul 2019, citing IQVIA

WHY NOW

Money became programmable. Now it can answer to evidence.

$

Paxos USDG

A US-dollar stablecoin from Paxos Digital Singapore. The only money CargoFlow moves.

RH

Robinhood Chain

An Arbitrum chain built for tokenised real-world assets, with USDG deployed.

P-256

Passkeys on chain

The RIP-7212 precompile lets a buyer pay with Face ID, no seed phrase.

Rs

Arbitrum Stylus

Evidence fusion in Rust, measured 19x cheaper than the Solidity reference.

USDG 0x7E95...802F on chain 46630 · 10 contracts source-verified · Stylus: 31,511 vs 611,945 gas at 128 readings (stylus/README.md)

THE SOLUTION

Capital that answers to the cargo’s evidence.

1. A financier escrows USDG for one shipment.
2. Tranches release only when signed sensor evidence passes an on-chain policy.
3. The buyer's one payment repays the financier, pays the exporter and hands over the bill of lading.



Illustrative characters. Everything they do is a real contract call.

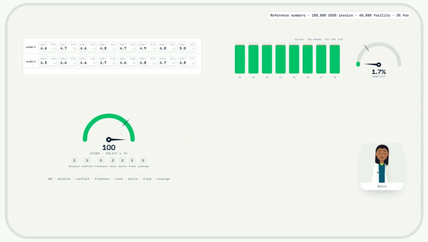
HOW IT WORKS

Five steps, every one on chain.



1 Escrow

Daniel deposits the 40,000 USDG facility into the vault.



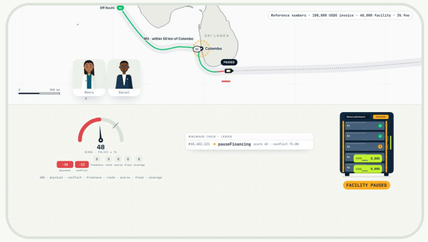
2 Evidence

Signed readings fuse into a score from 0 to 100.



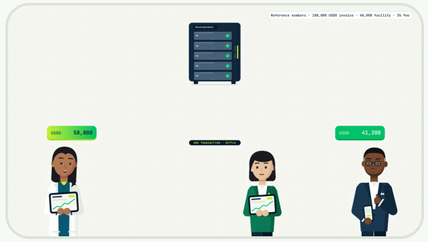
3 Release

Evidence passes the policy: 8,000 USDG to Meera.



4 Pause, prove

Score 48 pauses it; a Groth16 proof resumes it.



5 Settle

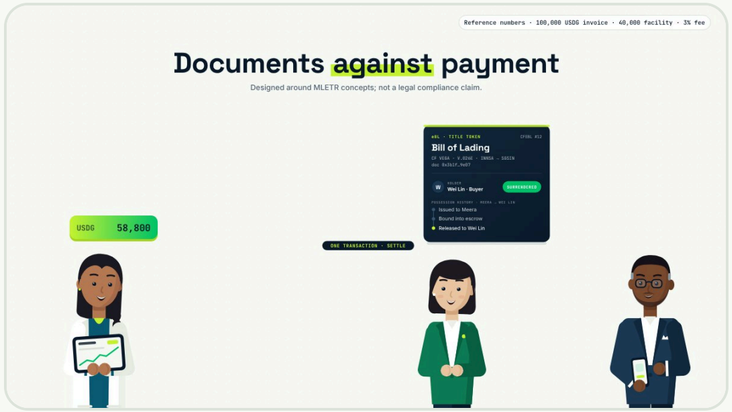
One 100,000 USDG payment splits and moves the title.

Reference numbers: 100,000 USDG invoice, 40,000 USDG facility, 3% fee.

createFacility · depositCapital · commitEpoch · evaluateAndReleaseMilestone · pauseFinancing · resumeWithProof · settle (contracts v3, 3 Oct 2026)

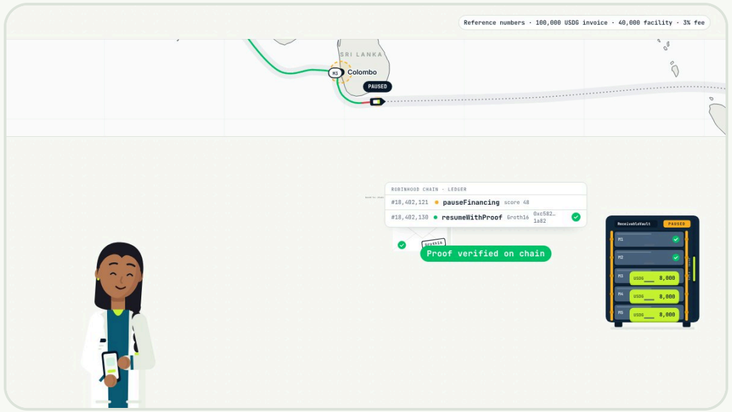
BUILT INTO THE CONTRACTS

Safety nets are part of the protocol.



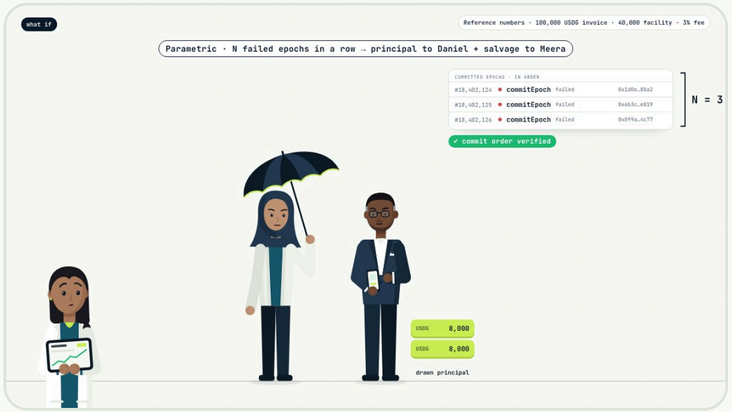
Title moves with money

The carrier's ERC-721 bill of lading leaves escrow inside the buyer's payment.



Automatic recovery

A proof is prepared once a probe has eight fresh in-band readings. The exporter signs once.

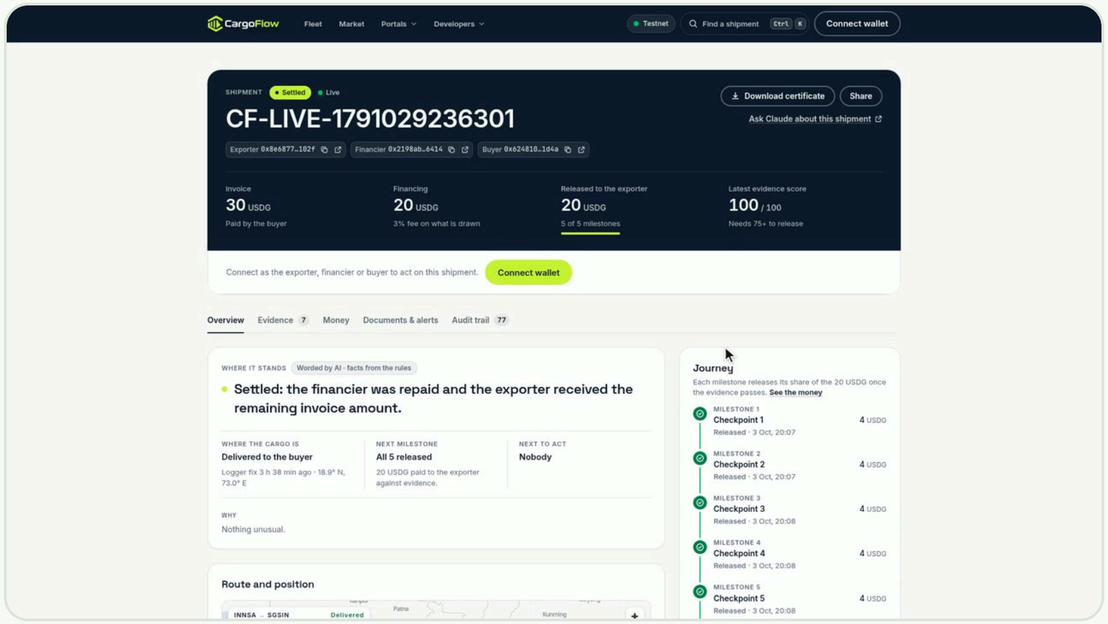


Parametric cover

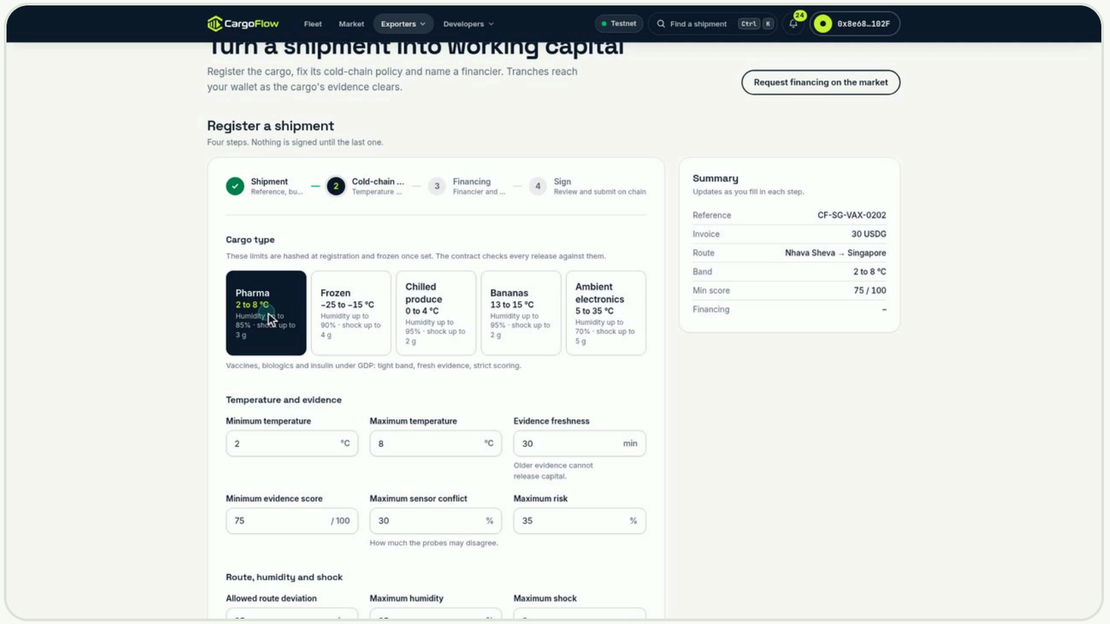
Pays out after N failed epochs in a row, proven from the commit order.

LIVE AT CARGOFLOW.ADORANTO737.WORKERS.DEV

A product each party can use today.



Shipment dashboard: settled, five of five milestones released.

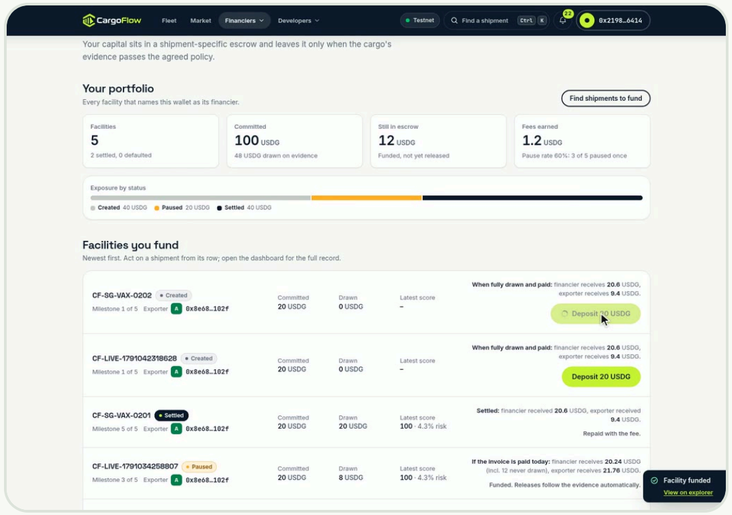


Exporter wizard: the Pharma 2 to 8 °C template fills the policy.

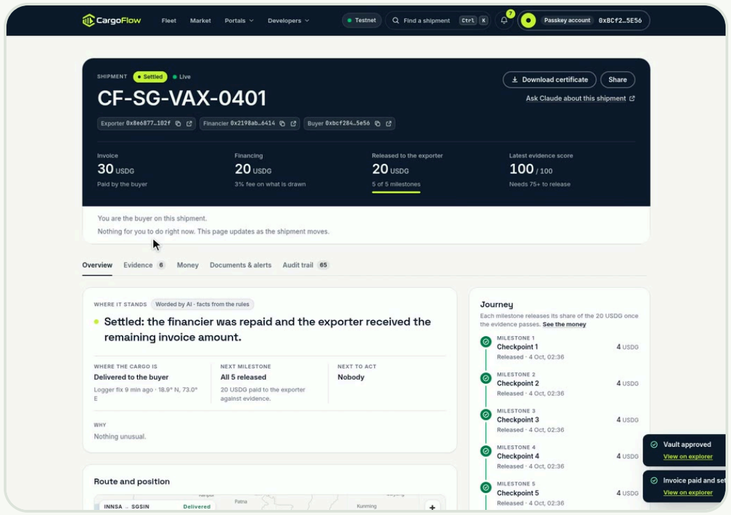
Pages: /track, /exporter, /financier, /buyer, /market, /ebl, /arbiter, /shipments, /developers, /docs, /deployments · open the live app

EVERY PARTY, ITS OWN SESSION

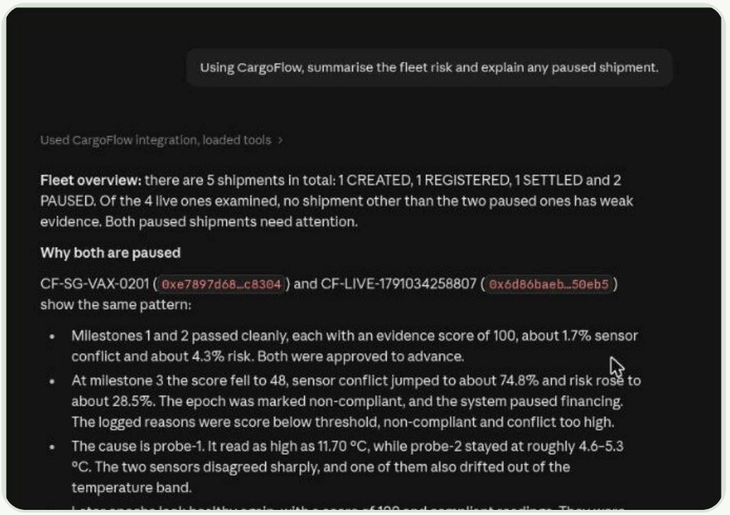
Fund it, pay it, ask Claude about it.



Financier: portfolio, exposure and one-click deposits.



Buyer: confirms delivery and pays with a passkey.



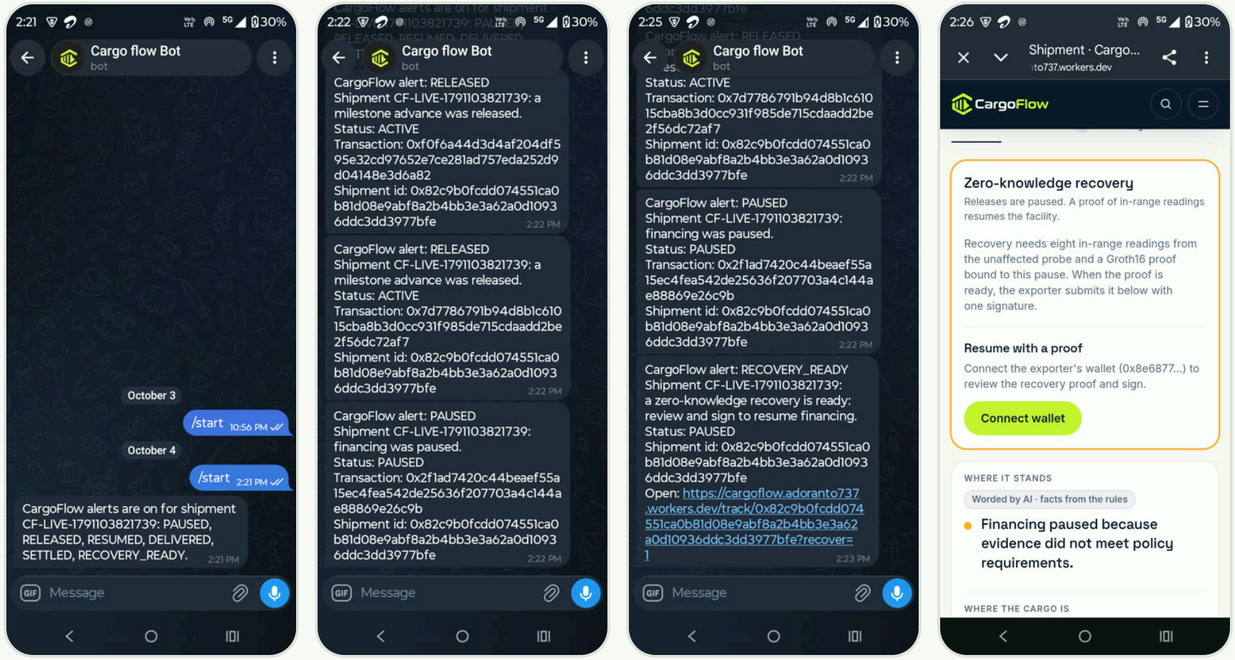
claude.ai: 25 tools through remote MCP. Claude never holds a key.

Remote MCP server: cargoflow-mcp.adoranto737.workers.dev/mcp · every write is signed by the party in its own wallet or passkey

ALERTS

Alerts on Telegram, email, Slack and webhooks.

Recorded on a real phone during a live testnet run. RELEASED and PAUSED arrive within seconds of the chain events; RECOVERY_READY opens the recovery card, ready to sign.



Alerts on

Released, paused

Recovery ready

One tap to sign

Bot: @Cargo_FlowBot

Shipment CF-LIVE-1791103821739 · pause 0x2f1ad742… · releases 0xf0f6a44d… and 0x7d778679… · also in-app

TECHNOLOGY AND MOAT

Five mechanisms that enforce each other.

01

Signed readings

Each device signs; every epoch records its source.

02

Fusion score

Dempster-Shafer fusion with a published formula.

03

Bound ZK proof

Groth16, bound to the contract's own signals.

04

Locked vault

Only the controller's rules release USDG.

05

Title with payment

The bill of lading moves inside the payment.

A copy needs all five, tested together: **366** contract tests and **25** circuit tests.

Groth16: 13,494 constraints, about 1 s to prove, about 0.25 M gas to verify · immutable core, eight fuzzed invariants · docs/architecture.md, docs/benchmarks.md

TRACTION

Production build live on Robinhood Chain Testnet.

Mainnet release next.

10
contracts source-verified on chain

4
packages published on npm and PyPI

25
MCP tools on a public remote server

3
extensions on Arbitrum Sepolia

TESTS, ALL PASSING

366	25	296	18	92	27	36	25
contract	circuit	frontend	end-to-end	SDK	MCP	gateway	Python

@cargoflow/sdk, @cargoflow/mcp, @cargoflow/gateway on npm and cargoflow on PyPI (0.1.0) · Fhenix and GMX Sourcify-verified, Stylus engine · /deployments

LIVE RUNS

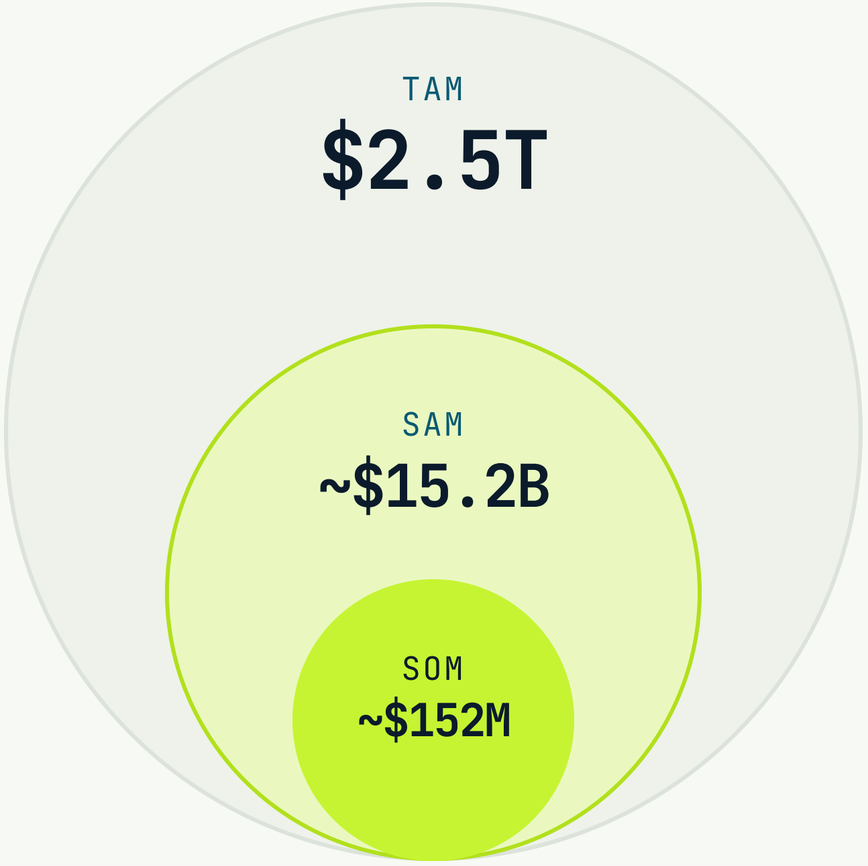
Full lifecycles settled on chain, with receipts.

CF-SG-VAX-0202	Website run: excursion, automatic ZK recovery, settlement.	proof 0x79c68fc0… settle 0xabba66b6…
CF-SG-VAX-0401	The buyer paid with a passkey: three ERC-4337 user operations.	account 0xBCf28437… settle 0xee1d5ba8…
CF-LIVE-1791029236301	v3 reference run through the hosted API, each party signing.	proof 0xadfe3b2f… paid 0x37571b49…
v1 hero run	22 transactions in 116 seconds, proof verified on chain.	proof 0x3910c2a5… settle 0xcb11761c…

Every hash links to explorer.testnet.chain.robinhood.com · full transaction tables in the README, "Live on Robinhood Chain Testnet"

MARKET SIZE, BOTTOM-UP

Start with one corridor inside a $2.5T gap.



TAM
$2.5T

Unmet trade finance demand in 2025 (ADB).

SAM
~$15.2B

India's pharma exports, $30.47B, × 50% sold on credit. A1

SOM
~$152M

1% of SAM financed through CargoFlow by year 3. A2

REVENUE
$0.46–0.76M

A planned 30 to 50 bps fee on that volume, per year. A3

CAPITAL
~$22M

Financier capital outstanding at 52-day terms. A4

Assumptions: A1 Atradius' India-wide B2B credit share applies to pharma · A2 1% is a target · A3 planned fee level · A4 365 / 52, about 7 turns a year · PIB, FY 2024-25 pharma exports · ADB · Atradius

BUSINESS MODEL

Paid when capital moves, not before.

01

Protocol fee

30 to 50 bps of each drawn facility, inside the settlement waterfall (A3).

02

Cover premiums

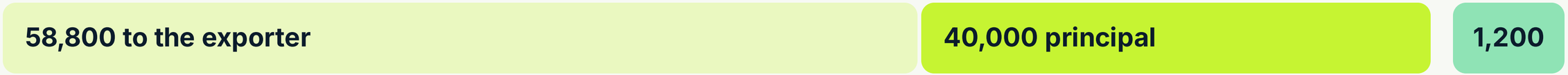
A share of default and parametric cover written through the CoverPool.

03

Software and data

Portfolio risk, evidence API and MCP for funds and insurers.

ONE 100,000 USDG PAYMENT, SPLIT ON CHAIN (REFERENCE FACILITY)



Principal and the 3% fee go to the financier. The protocol fee is one more line in this split.

Non-custodial: the ReceivableVault is the sole USDG custodian · fee levels are planned (A3) · contracts/src/ReceivableVault.sol, CoverPool.sol, packages/python

LANDSCAPE, BY CATEGORY

Each category solves one piece. CargoFlow joins them.

	Sees cargo condition	Releases on evidence	Payment and title together	Private telemetry	Stablecoin settlement
Bank trade finance letters of credit, collections	○	◐	◐	○	○
Factoring fintechs advance on the buyer's credit	○	○	○	○	○
Cold-chain monitoring loggers and visibility	●	○	○	○	○
On-chain RWA credit pooled private credit	○	○	○	○	●
CargoFlow a facility per shipment	●	●	●	●	●

● yes ◐ partly, on documents ○ not a design goal of the category

Categories described by their typical design, not by any one company. Ratings are CargoFlow's assessment; individual providers may differ.

GO-TO-MARKET

One corridor first: India pharma to Singapore.



Nhava Sheva — Colombo — Singapore

DOOR 1 · LOGGER VENDORS

Make existing loggers finance-ready

@cargoflow/gateway signs and sends logger exports.

DOOR 2 · CREDIT FUNDS

Short-tenor USDG yield, evidence attached

The market suggests a fee band per request.

DOOR 3 · FORWARDERS, INSURERS

Reach exporters through who they use

The Pharma template sets 2 to 8 °C in one click.

Pilot target: 3 exporters, 1 financier and 1 logger vendor on this lane, then more lanes and cold-chain categories · India pharma exports $30.47B in FY 2024-25 (PIB)

ROADMAP, EACH MILESTONE WITH AN EXIT TEST

From production build to the first real facility.

M0 done: contracts v3 verified, live app, API and MCP, passkey payments, ZK recovery, four packages.

M1

Security

Independent audit and a public ZK ceremony

35% OF FUNDING

M2

Real evidence

Logger pilots and a secure-element board

25% OF FUNDING

M3

Mainnet

Real USDG on Robinhood Chain, protocol fee

20% OF FUNDING

M4

Legal and title

MLETR opinion on the bill of lading

10% OF FUNDING

M5

Scale

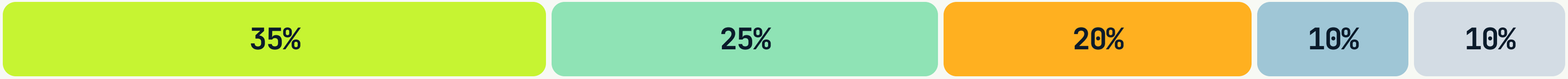
Calibrated scoring, a second corridor

AFTER THIS ROUND

Exit tests: audit report published · 3 real shipments evidenced from physical loggers · first outside-financier facility settled in real USDG · written legal opinion · second corridor live

THE ASK: MILESTONE-RELEASED FUNDING

Prize and seed money fund M1 to M4.



Security audit and public ZK ceremony	M1	**35% of funds**
Pilots and logger hardware	M2	**25% of funds**
Mainnet launch and first facilities	M3	**20% of funds**
Legal: MLETR opinion and structuring	M4	**10% of funds**
Team and infrastructure, six months	ALL	**10% of funds**

Released like a CargoFlow facility: each tranche follows a milestone's exit test.

A plan, not a commitment: shares of whatever the round provides, whether prize, accelerator or seed money · figures are percentages on purpose

THE ROBINHOOD CHAIN TRACK

What is live, with something to click.

Robinhood Chain

10 contracts verified on Robinhood Chain Testnet (46630).

Paxos USDG

The only money CargoFlow moves.

Working product

Five portals, an API, an MCP server, four packages.

Real runs

End-to-end runs with every transaction linked.

Chain features

Its RIP-7212 precompile carries the passkey flow.

Arbitrum

Stylus, Fhenix and GMX on Arbitrum Sepolia.

Contract quality

366 tests, eight invariants, an immutable core with no proxy.

Read the Crypto World's Fair announcement

Evidence: README "Built on Robinhood Chain", "Live on Robinhood Chain Testnet", "Measured, not claimed" · docs/project/21-hackathon-alignment.md

TEAM AND CONTACT

Built by its founder, live and open.



Arpit Singh

Founder

github.com/LSUDOKO

WEBSITE	cargoflow.adoranto737.workers.dev
GITHUB	github.com/LSUDOKO/CargoFlow
NPM	@cargoflow/sdk · mcp · gateway
PYPI	pypi.org/project/cargoflow
CLAUDE	cargoflow-mcp.adoranto737.workers.dev/mcp